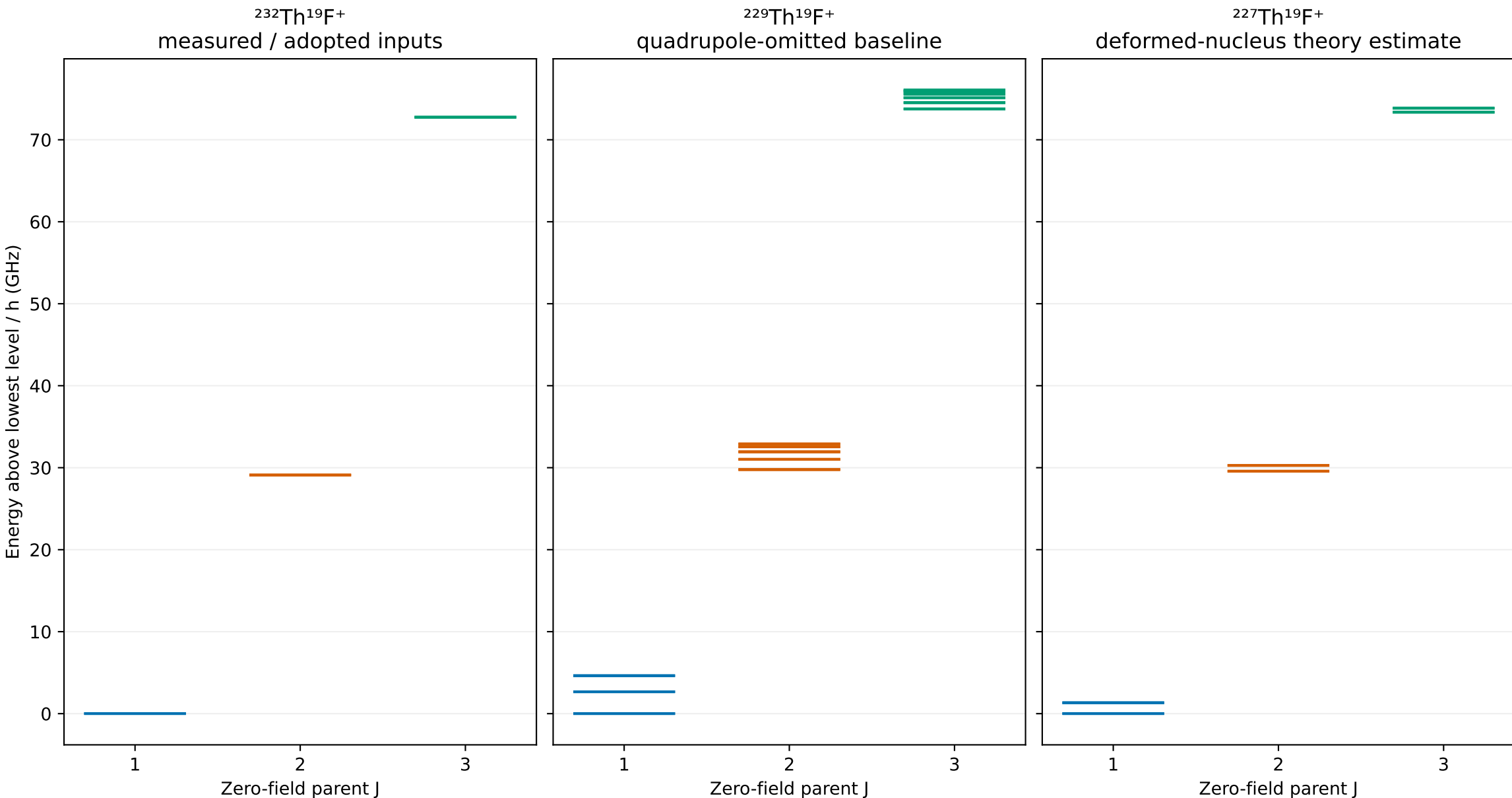
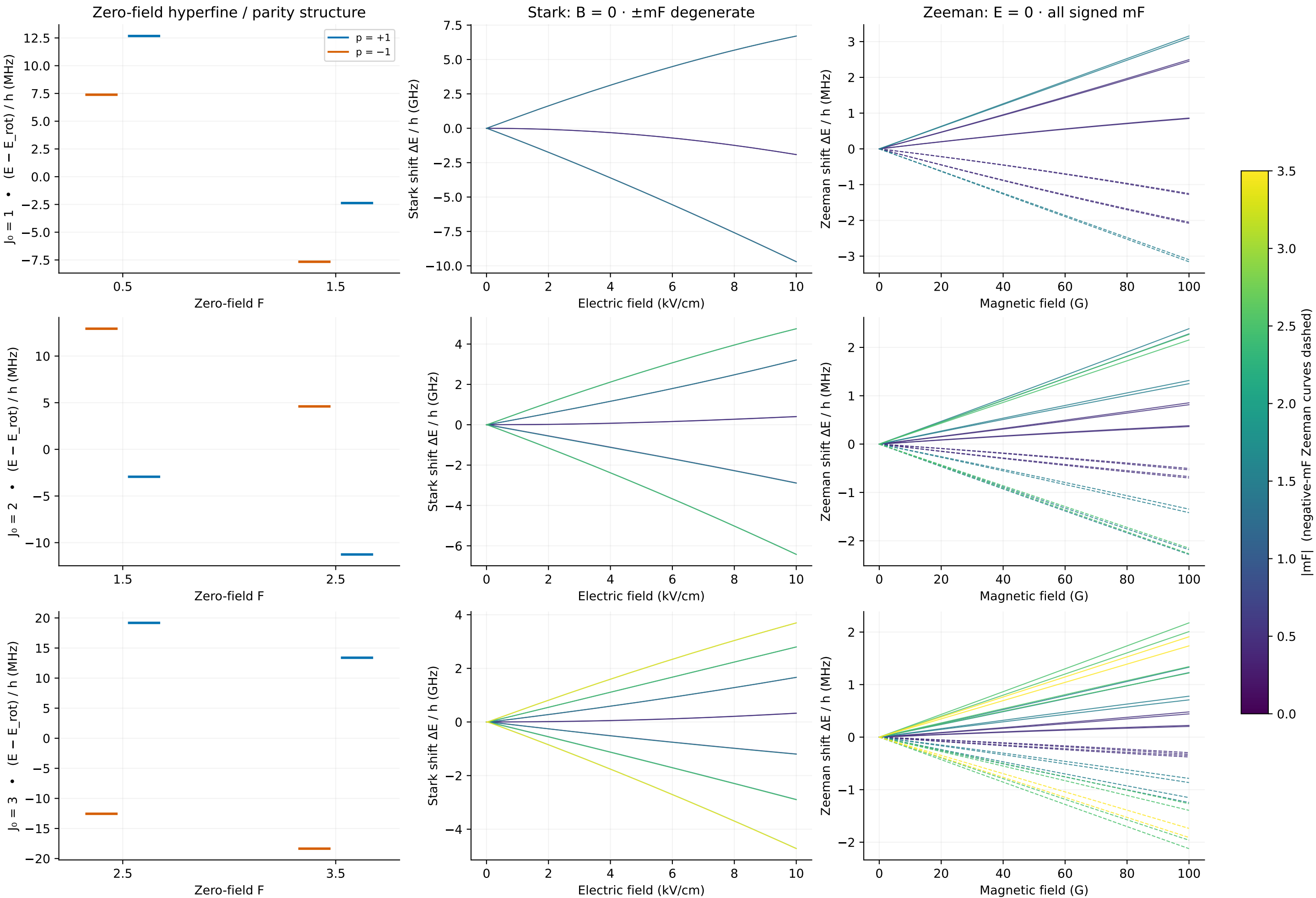


ThF⁺ X ³Δ₁ · zero-field rotational structure



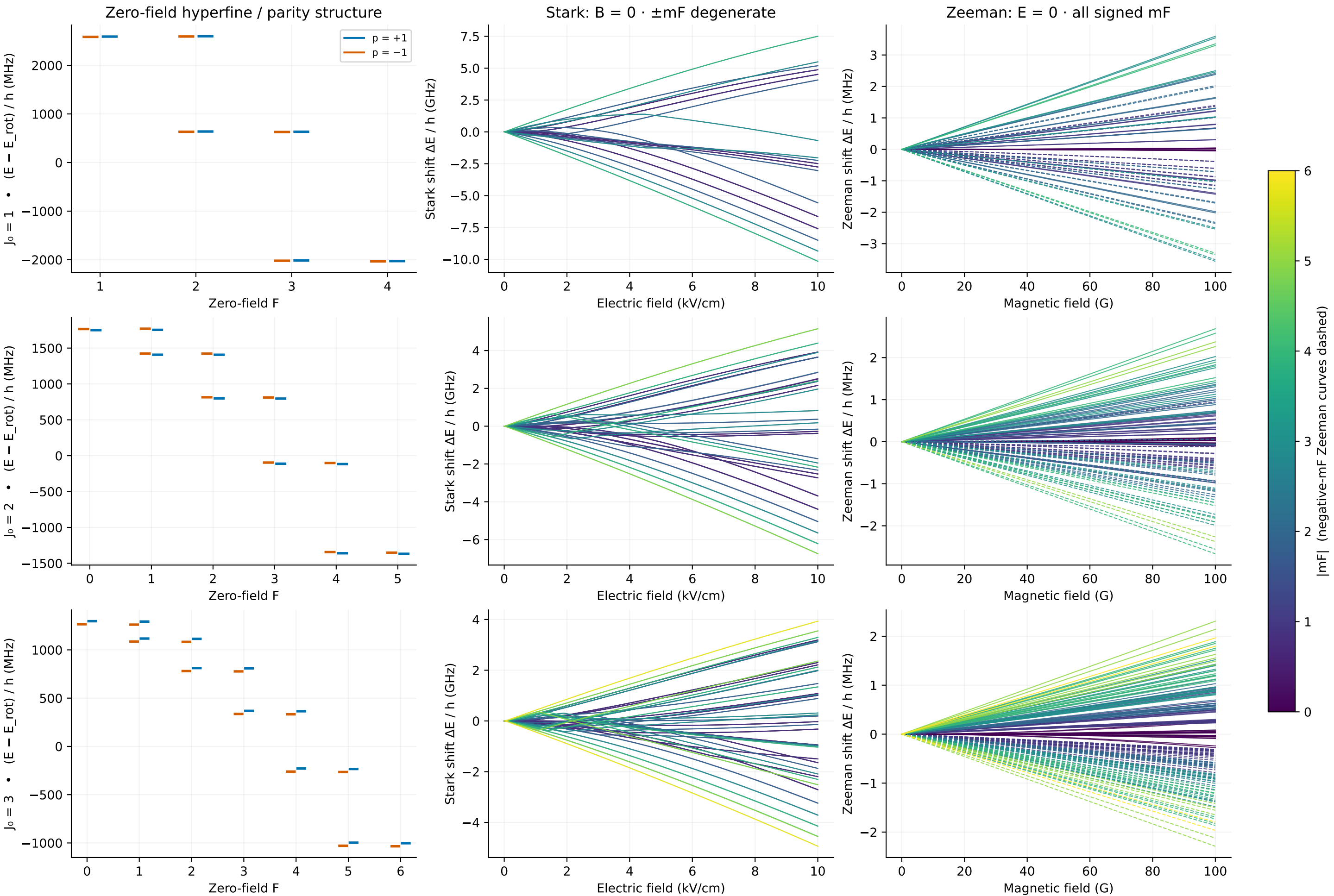
Each isotope has its own zero. Odd-isotope molecular constants are unscaled transfers from ²³²ThF⁺.
 Unknown Th spin rotation omitted; ²²⁹Th quadrupole omitted. Exploratory models without physical uncertainty bands.

$^{232}\text{Th}^{19}\text{F}^+$ · measured / adopted inputs
 $X^3\Delta_1$ · levels correlated with $J = 1-3$ · basis $J \leq 8$



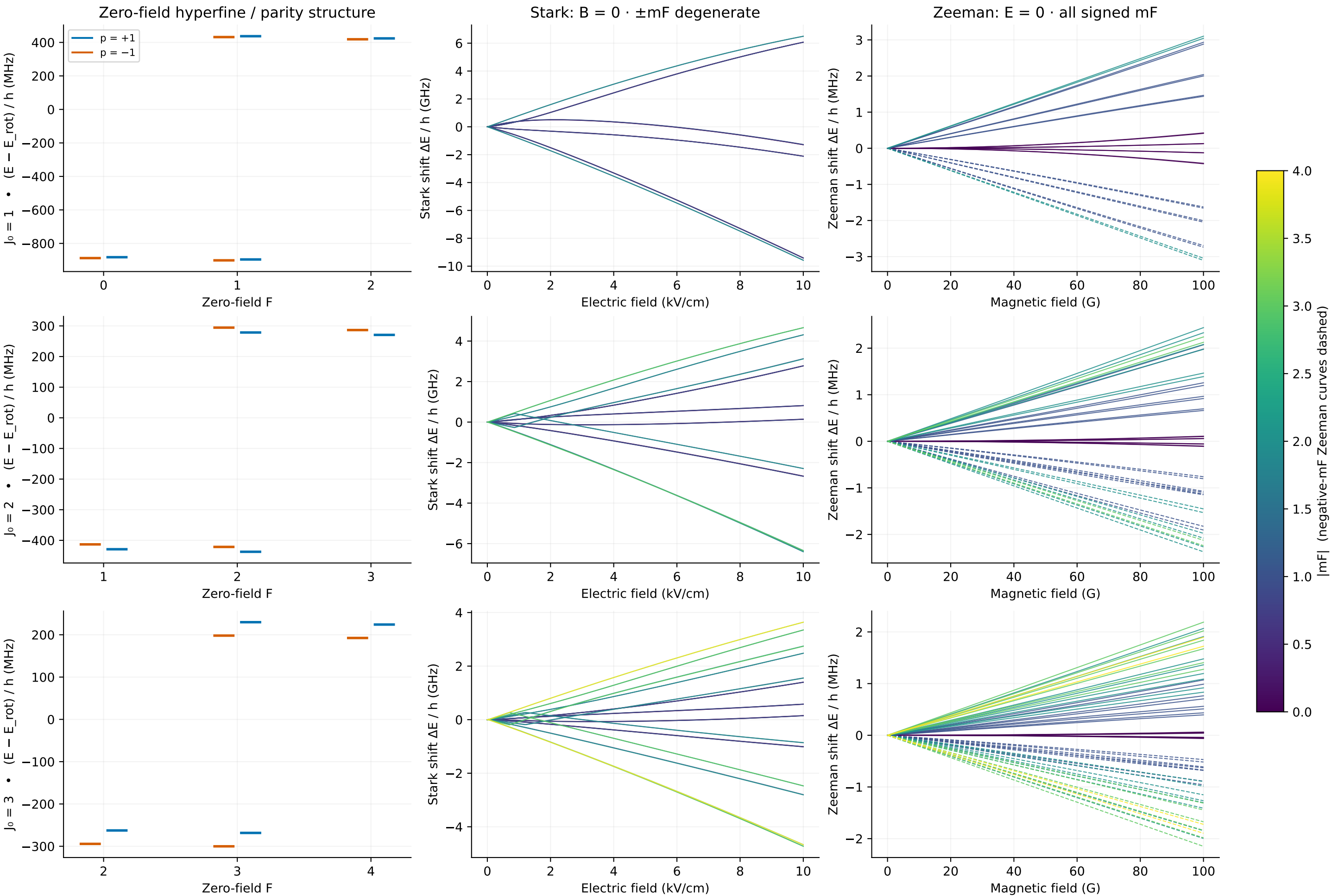
Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).

$^{229}\text{Th}^{19}\text{F}^+$ · quadrupole-omitted baseline
 $X^3\Delta_1$ · levels correlated with $J = 1-3$ · basis $J \leq 8$



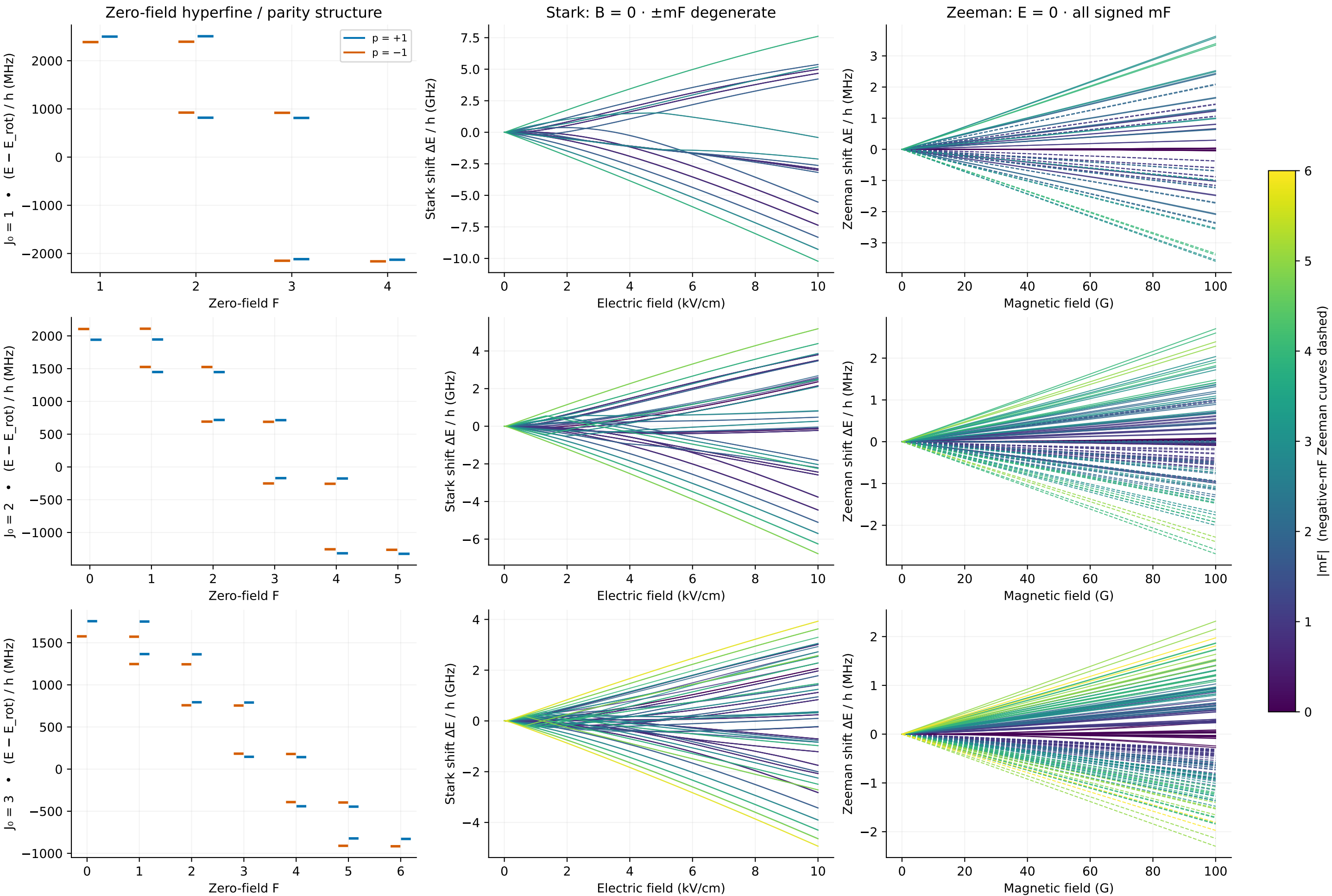
Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF, p).
Molecular constants transferred unscaled from $^{232}\text{ThF}^+$; unknown Th spin rotation omitted; no physical uncertainty band.

$^{227}\text{Th}^{19}\text{F}^+$ · deformed-nucleus theory estimate
 $X^3\Delta_1$ · levels correlated with $J = 1-3$ · basis $J \leq 8$



Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF, p).
Molecular constants transferred unscaled from $^{232}\text{ThF}^+$; unknown Th spin rotation omitted; no physical uncertainty band.

$^{229}\text{Th}^{19}\text{F}^+$ · UNVALIDATED quadrupole sensitivity case
 $X^3\Delta_1$ · levels correlated with $J = 1-3$ · basis $J \leq 8$



Exploratory model; uncertainty not shown. Zero-field J labels; J mixes in E. Stark: stepwise character tracking; Zeeman: energy order within (mF,p).
Molecular constants transferred unscaled from $^{232}\text{ThF}^+$; unknown Th spin rotation omitted; no physical uncertainty band.